Md Tanvir Rahman Sahed

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EDUCATION

BSc. in Electrical, Electronic and Communication Engineering (EECE) | Feb 2020 – Apr 2024
@MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY (MIST) , CGPA: **3.49/4.00** | Courses

INDUSTRY EXPERIENCE

Embedded systems Engineer | *Jun 2025 – Oct 2025*
@Creative Factory Labs (CFL)

- Designed custom embedded **firmware and PCB** systems for a **German automotive** client, ensuring real-time reliability and hardware–software co-design.
- Engineered a portable, modular, and scalable firmware architecture with HAL abstraction and interrupt-driven **control for safety-critical** applications.
- Delivered **industrial-grade systems**, including a hydropower plant valve motor controller and an ECU ignition control unit with precise timing regulation.

R&D Engineer, Embedded systems and Robotics | *Oct 2024 - Jun 2025*
@Electrical Central Lab, NRG Composite yarn Dyeing, NR Group

- Developed 5+ **microcontroller-based automation** systems integrated with 3D mechanical design, reducing manual intervention by 40%.
- Implemented **industrial sensor communication** with closed-loop controls, improving stability by 25%.
- Delivered 3 functional **prototypes**, including a smart water flow meter (±2% accuracy) and an automated laboratory dyeing machine with temperature control within ±1 °C tolerance.

RESEARCH EXPERIENCE

Graduate Research Assistant | *Oct 2025 - Current*
@Smart Grid and Sustainable Energy Transition Lab (SETL) , EECE, MIST

- Funded project to Develop an **AI enabled Digital Twin** for solar power plant by Bangladesh Energy and Power Research Council(BEPRC) .
- Building a data-driven Digital Twin for plant-state replication and decision support using **time-series forcasting, reinforcement learning**.
- Data analysis from **Embedded device and sensor network** of the plant to do predictive maintenance.

Student Researcher | *Apr 2023 - Apr 2024*
@Visual Image Processing Research Lab (VIPL) , EECE, MIST

- Bridged the low-resource gap in **Bengali visual speech recognition** by processing **1400+ hours** of video data and pioneering a dataset spanning 150 speakers and 500 annotated words.
- Developed a **3D CNN + BiLSTM + CTC** visual speech recognition pipeline for spatiotemporal feature extraction and sequence modeling.
- Achieved **84.6%** recognition accuracy and published the work as first author in *Data in Brief* (Elsevier).

PUBLICATIONS (google scholar)

LipBengal: Pioneering Bengali lip-reading dataset for pronunciation mapping through lip gesture | 2025
Sahed, M.T.R., Aronno, M.T.I., Nyeem, ...@**Vol 58, *Data in Brief*, Elsevier, 2025**. DOI: 10.1016/j.dib.2024.111254

Object Classification by a 7-DOF Robotic Arm Using MobileNet-SSD and Feedback Position Mapping | 2025
Md Tanvir Rahman Sahed, Shoaib Rufsun, ...@**IEE QPAIN**. DOI: 10.1109/QPAIN66474.2025.11171666

SKILLS

Technical Skill Sets

Programming Language: C/C++, Python, MATLAB, Verilog

Embedded & Real-Time Systems: AVR, ESP, and ARM Cortex-M MCUs || firmware architecture (modular, interrupt-driven, state machines, etc.) || communication protocols (Modbus, I²C, MQTT,etc.) || real-time scheduling || low-level driver development || debugging.

Machine Learning & Deep Learning: CNN and RNN architectures, object detection (YOLO, MobileNet-SSD), classical ML (SVM, kNN, Random Forest, Gradient Boosting, etc.), cross-validation, and performance benchmarking.

Hardware & Circuits: SPICE-based circuit simulation, schematic capture and PCB (EasyEDA), power/signal conditioning, board bring-up, debugging, and validation testing.

Design: Parametric 3D modeling (Fusion 360) and printing, VLSI design (Cadence Virtuoso)

PROJECTS

Ignition System for Custom Engine Control Unit (STM32F4xx)

Designed a real-time ignition control hardware and firmware with encoder-based RPM tracking and dynamic dwell angle computation for multi-cylinder engines, written in C inside STM32 Cube IDE for Electric Vehicles... learn more

Hydropower Plant Valve Motor Controller

Developed a dual Modbus-enabled STM32F4xx motor control system for precise valve actuation and industrial automation integration, ARM Cortex-M processor and secondary network chip, shadow copy NVS sys... learn more

MCU-Powered Lab Dyeing Machine

Engineered a microcontroller-based closed-loop temperature and motor control system for a sample dyeing machine using avr processor and secondary devices at 20% the original cost, achieving 0.1% tolerance in tem... learn more

Smart Water Flow Meter

AVR MCU-powered and HAll effect sensor-based water consumption tracker with QT (Python)-based GUI, Modbus communication and data logging support. This purpose is to automate water treatment plant to save... learn more

KRAKEN Foam-Collecting Robot Chassis

Designed a amphibious robotic chassis to collect accumulating foam from refined wastewater in an efflu... 3D view

COMPETITIONS & CONFERENCE

Competitions and Achievements

- Semi-finalist, Line Follower Robot Competition, Techfest IIT Regional (IUBAT), 2022.Credential | Event
- Team Lead, team of 4 in the Soccer Bot Competition, EEE Day (KUET), 2023.Credential | Event
- Technical Supervisor, 3 teams in ESONANCE (IUT), 2023.Credential | Event

Conference and Workshops

- Co-supervisor & Author, IEEE 2nd International Conference on Computing, Application and Systems (COMPAS), 2025. Article | Credential
- Attendee, Movers Workshop on SDG13: Climate Action, Level 2.Credential

EXTRA-CURRICULAR AFFILIATIONS

President, Robotics Club MIST | Apr 2023 - Feb 2024

A student-led organization where I led a team of 8, mentored others, and organized workshops. Credential | Club